

EXPERIMENT - 9

Date: 2022-11-6

TO VERIFY THE PRINCIPLE OF MOMENTS AND HENCE DETERMINE THE MASS OF GIVEN UNKNOWN BODY

APPARATUS REQUIRED:

- i. A meter scale
- ii. Slotted weights
- iii. Twin thread
- iv. Unknown body

THEORY:

The principle of Moments states that "if a body is in rotational equilibrium, the sum of the clockwise moments is equal to the sum of the anticlockwise moments". This principle may be utilized to find the magnitude and direction of unknown

force acting on a body present in equilibrium.

If P and Q be the known forces acting at points A and B, as shown in figure, then according to the principle of moments:

$$P \times OA = Q \times OB$$

By using this relation principle of moments can be verified.

Again, if W be the unknown load and W_1 the known load acting at points C and D respectively in such a way that the body is still in equilibrium condition, then

$$W \times OC = W_1 \times OD$$

$$W = \frac{OD}{OC} \times W_1$$

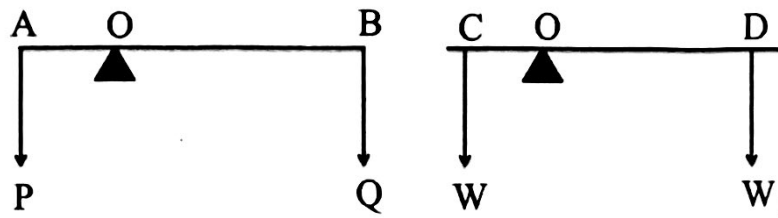


Fig. (i)

Fig. (ii)

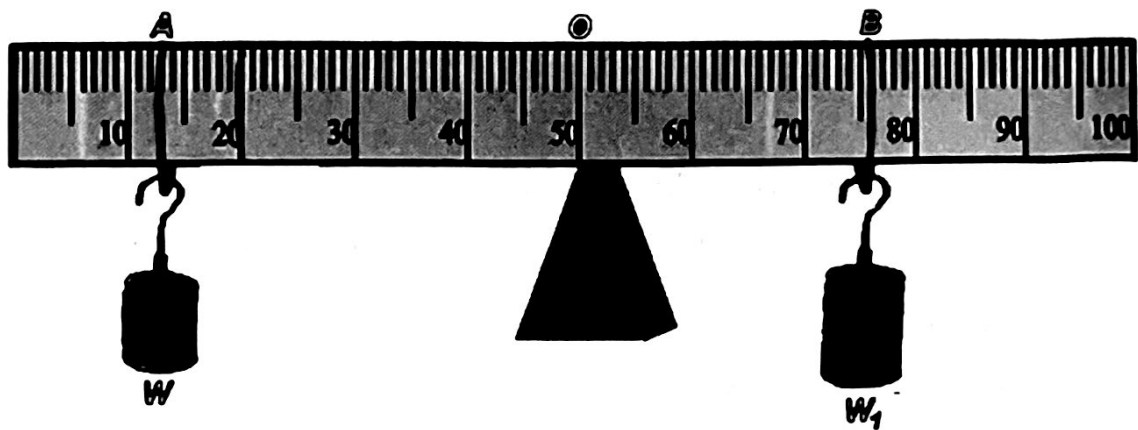


Fig. (iii)

OBSERVATIONS:L.C. of scale = 0.1 cmPosition of C.G. = 50.4 cm**Table 1: For the verification of principle of moment (Fig. i)**

No. of obs.	Weight on the		Distance from C.G.		moments		Remarks
	Left of C.G. P in gm	Right of C.G. Q in gm	OA Cm	OB Cm	Anticlockwise (P × OA)	clockwise (Q × OB)	
1	100	50	20.4	40.1	2040	2005	$P \times OA \approx Q \times OB$
2	200	100	20.4	40.2	4080	4020	$P \times OA \approx Q \times OB$
3	300	150	20.4	40.4	8120	6060	$P \times OA \approx Q \times OB$
4	400	200	20.4	40.7	8160	8140	$P \times OA \approx Q \times OB$
5	500	250	20.4	40.8	10200	10200	$P \times OA \approx Q \times OB$

Table 2: For the determination of unknown weight (Fig. ii)

No. of obs.	Weight On the		Distance from C.G.		Moments		Unknown mass W = $\frac{W_1 \times OD}{OC}$ (g)	Mean (g)
	Left of C.G. W	Right of C.G. W ₁	OC	OD	Anticlockwise (W × OC)	clockwise (W ₁ × OD)		
1	W	50	12.4	9.6	12.4W	480	38.7	38.91
2	W	100	12.4	4.9	12.4W	490	39.1	
3	W	150	12.4	3.2	12.4W	480	38.7	
4	W	200	43.4	8.6	43.4W	1720	39.2	
5	W	250	32.2	5	32.2W	1250	38.9	

RESULT:

Hence, we obtained the mass of unknown ball which was found out to be 38.92 g.

PERCENTAGE ERROR:Mass of the body = 39 g (measured by digital balance)Observed mass of the body = 38.925 g (from table: 2)

$$\therefore \% \text{ error} = \left| \frac{\text{S.V.} - \text{O.V.}}{\text{S.V.}} \right| \times 100\% = \underline{0.19} \%$$

CONCLUSION:

From the following experiment we verified the principle of moment and calculate the mass of unknown ball.

SOURCES OF ERROR:

- ① Inaccurately determined C.G. of the meter scale.
- ② Inaccurately positioning of thread on the scale at C.G. while doing experiment
- ③ non uniform mass distribution of meter scale
- ④ non uniform mass distribution of threads used for making loops.

PRECAUTIONS:

- ① The meter scale must have uniform mass distribution.
- ② In equilibrium position, the meter scale should be exactly horizontal.
- ③ Threads used for loops should be thin, light and strong. Their loops must be parallel to the marks on the scale and of equal lengths.

VIVA QUESTIONS:

1. Define moment of force.
2. State the principle of moments.
3. What is couple?
4. Why handles are fixed at free sides of the door?
5. Differentiate between center of gravity and center of mass.
6. Why is it danger to travel in a double Decker bus?
7. What are the like and unlike parallel forces? Explain.
8. What do you mean by clockwise and anticlockwise moments?
9. Can two unequal forces acting at a point be in equilibrium? Can three?
10. What is the difference rest and equilibrium?

1. The turning effect of force about a given axis of rotation is called moment of force. It is measured as the product of magnitude of force and perpendicular distance of force from given axis of rotation.

2. Principle of moment states that if a body is in equilibrium under the action of number of forces then sum of the clock wise moments is equal to the sum of anti-clock wise moments.

3. Two equal and opposite parallel forces acting on a body at different points is called couple. A couple produce pure rotational motion without translation.

4. Handles are fixed at the edge furthest from the hinges to maximize the perpendicular distance from the axis of rotation. This increases the moment of force, making it easier to open the door with less effort.

5. Center of Mass

Center of Gravity

- The unique point where the entire mass of an object may be considered to be concentrated.

The point through which the entire weight of the body acts. In a uniform gravitational field, these two points coincide.

6. A double-decker bus has a high center of gravity. When it tilts or turns sharply, the vertical line passing through its center of gravity can easily fall outside its base, causing the bus to become unstable and tip over.

7. Like parallel Forces

- Forces that act in the same direction along parallel lines.

• Unlike Parallel Forces

- Forces that act in opposite directions along parallel lines.

8. A moment that tends to rotate an object in the direction of clock hands, is called clockwise moment. By convention, it is often taken as negative.

- A moment that tends to rotate an object opposite to clock hands, is called Anticlockwise moment. By convention, it is often taken as positive.

9. No. To be in equilibrium two forces must be equal in magnitude and opposite in direction.

- Yes, if the resultant of any two forces is equal and opposite to the third force (forming a closed triangle of vectors).

10. Rest - A state where an object's velocity is zero relative to a frame of reference.

Equilibrium - A state where the net force and net torque acting on an equilibrium while moving at a constant velocity (dynamic equilibrium) or while at rest (static equilibrium).

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